

MAT 102: Ordinary Differential Equations

Topic 3: Second Order Linear Ordinary Differential Equations

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Table of contents

1	Overview	1
2	General Theory of Second Order Linear ODEs	2
3	Existence and Uniqueness Theorem	4
4	Linear Independence and the Wronskian	8
4.1	Linear Independence	8
4.2	The Wronskian	9
4.2.1	Examples using the Wronskian	9

1 Overview

Many things in nature move, vibrate, or change with time.

Examples include:

- a swinging pendulum,
- a vibrating spring,
- and electrical circuits.

These systems can often be described using **second-order linear differential equations**.

A **second-order linear ODE** involves the second derivative y'' as the highest derivative.

2 General Theory of Second Order Linear ODEs

The **general second-order linear ODE** has the form:

$$y'' + P(x)y' + Q(x)y = R(x) \quad (3.1)$$

where $P(x)$, $Q(x)$, and $R(x)$ are functions of x alone.

Equation (3.1) is called **linear** because y, y' , and y'' appear only to the first power.

- **Homogeneous** equation: $R(x) = 0 \Rightarrow y'' + P(x)y' + Q(x)y = 0$
- **Non-homogeneous** equation: $R(x) \neq 0$

The study of the general theory focuses on understanding:

- the structure of solutions,
- existence and uniqueness of solutions,
- homogeneous and nonhomogeneous equations,
- methods for finding solutions, etc

One important property of linear equations is the **superposition principle**, which states that linear combinations of solutions of the homogeneous equation are also solutions.

Superposition Principle

If (y_1) and (y_2) are solutions of the homogeneous equation

$$a(x)y'' + b(x)y' + c(x)y = 0,$$

then any linear combination

$$y = C_1y_1 + C_2y_2$$

is also a solution, where (C_1) and (C_2) are constants.

Proof

Since (y_1) and (y_2) are solutions,

$$a(x)y_1'' + b(x)y_1' + c(x)y_1 = 0$$

and

$$a(x)y_2'' + b(x)y_2' + c(x)y_2 = 0.$$

Let

$$y = C_1y_1 + C_2y_2.$$

Differentiate:

$$y' = C_1y_1' + C_2y_2'$$

and

$$y'' = C_1y_1'' + C_2y_2''.$$

Substitute into the differential equation:

$$\begin{aligned} & a(x)y'' + b(x)y' + c(x)y \\ &= a(x)(C_1y_1'' + C_2y_2'') + b(x)(C_1y_1' + C_2y_2') + c(x)(C_1y_1 + C_2y_2). \end{aligned}$$

Group the terms:

$$= C_1[a(x)y_1'' + b(x)y_1' + c(x)y_1] + C_2[a(x)y_2'' + b(x)y_2' + c(x)y_2].$$

Since (y_1) and (y_2) satisfy the homogeneous equation,

$$a(x)y_1'' + b(x)y_1' + c(x)y_1 = 0$$

and

$$a(x)y_2'' + b(x)y_2' + c(x)y_2 = 0.$$

Therefore,

$$a(x)y'' + b(x)y' + c(x)y = C_1(0) + C_2(0) = 0.$$

Hence,

$$y = C_1y_1 + C_2y_2$$

is also a solution of the homogeneous equation. ■

In the rest sections, we develop the theory and methods used to solve second-order linear ODEs and study their applications to real-world problems.

3 Existence and Uniqueness Theorem

Theorem — Existence and Uniqueness

Consider the IVP:

$$y'' + P(x)y' + Q(x)y = R(x), \quad y(x_0) = y_0, \quad y'(x_0) = y_1$$

If $P(x)$, $Q(x)$, and $R(x)$ are **continuous** on an open interval I containing x_0 , then there exists a **unique** solution $y = \phi(x)$ defined on the **entire** interval I .

The interval of validity I is the largest interval containing x_0 where the coefficients remain continuous.

This theorem answers two important questions:

1. **Does a solution exist?**
2. **Is the solution unique?**

The theorem says:

- A solution **does exist**, and there is **only one** solution satisfying the given initial conditions, provided the coefficients are continuous.

Key points:

1. Why are two initial conditions needed?

A second-order differential equation contains the second derivative y'' , so two integrations are usually required to obtain y .

Each integration introduces a constant:

$$C_1, \quad C_2.$$

Therefore, two initial conditions are needed to determine these constants uniquely:

$$y(x_0) = y_0, \quad y'(x_0) = y_1.$$

2. Importance of continuity

The theorem requires that

$$P(x), \quad Q(x), \quad R(x)$$

be continuous on the interval.

If one of them is discontinuous at some point, the theorem no longer guarantees existence and uniqueness beyond that point.

Example 1: Use the Existence and Uniqueness Theorem to determine the interval on which the IVP has a unique solution.

$$y'' + (\tan x)y' + y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Here,

$$P(x) = \tan x, \quad Q(x) = 1, \quad R(x) = 0.$$

The function $\tan x$ is discontinuous at

$$x = \pm \frac{\pi}{2}.$$

Since the initial point is $(x_0=0)$, the largest interval containing (0) where all coefficients are continuous is $(-\frac{\pi}{2}, \frac{\pi}{2})$. Therefore, the theorem guarantees a unique solution on

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Example 2: Determine whether the following IVP has a unique solution near the initial point.

$$y'' + \frac{1}{x-2}y' + y = e^x, \quad y(3) = 0, \quad y'(3) = 1.$$

We identify

$$P(x) = \frac{1}{x-2}, \quad Q(x) = 1, \quad R(x) = e^x.$$

The function

$$\frac{1}{x-2}$$

is discontinuous at $x = 2$.

Therefore, by the Existence and Uniqueness Theorem, a unique solution exists on

$$(2, \infty).$$

Example 3 Find the largest interval containing the initial point for which the theorem guarantees a unique solution.

$$(x-4)y'' + y' + xy = \sin x, \quad y(5) = 2, \quad y'(5) = 0.$$

First write the equation in standard form by dividing by $x-4$:

$$y'' + \frac{1}{x-4}y' + \frac{x}{x-4}y = \frac{\sin x}{x-4}.$$

Thus,

$$P(x) = \frac{1}{x-4}, \quad Q(x) = \frac{x}{x-4}, \quad R(x) = \frac{\sin x}{x-4}.$$

All are discontinuous at

$$x = 4.$$

Since the initial point is $x_0 = 5$, the largest interval containing 5 is $(4, \infty)$. Hence a unique solution is guaranteed on

$$(4, \infty).$$

TRY

Q1: Use the Existence and Uniqueness Theorem to determine the interval on which the IVP has a unique solution.

$$y'' + x^2y' + (\cos x)y = \ln(x^2 + 1),$$

with

$$y(1) = 0, \quad y'(1) = 3.$$

Q2: Find the largest interval containing the initial point for which the theorem guarantees a unique solution.

$$y'' + \frac{1}{x-1}y' + y = e^x,$$

with

$$y(0) = 2, \quad y'(0) = 1.$$

4 Linear Independence and the Wronskian

When solving a second-order homogeneous differential equation,

$$y'' + P(x)y' + Q(x)y = 0,$$

we need **two independent solutions** to form the general solution.

For example,

$$y = C_1y_1 + C_2y_2.$$

But this only works if y_1 and y_2 are **linearly independent**.

4.1 Linear Independence

Definition

Two functions are linearly dependent if one is a constant multiple of the other:

$$y_2 = ky_1$$

for some constant k .

Otherwise, they are called **linearly independent**.

Simple Idea

- **Dependent** $\rightarrow$ same type of function, only multiplied by a constant.
- **Independent** $\rightarrow$ genuinely different functions.

Examples:

- $y_1 = e^{2x}$, $y_2 = 3e^{2x}$: **Dependent** ($y_2 = 3y_1$).
- $y_1 = e^{2x}$, $y_2 = e^{-2x}$: **Independent** (neither is a multiple of the other).
- $y_1 = \sin x$, $y_2 = \cos x$: **Independent**.
- $y_1 = x^2$, $y_2 = x|x|$: Tricky — check using the Wronskian.

4.2 The Wronskian

Definition — Wronskian

The **Wronskian** of two differentiable functions y_1 and y_2 is:

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

Wronskian Test

- If $W(x) \neq 0$ for some x in the same interval $\rightarrow$ **independent**
- If $W(x) = 0$ everywhere $\rightarrow$ **dependent**

4.2.1 Examples using the Wronskian

Example 1: Show $y_1 = e^{2x}$ and $y_2 = e^{-3x}$ are linearly independent.

$$W = \begin{vmatrix} e^{2x} & e^{-3x} \\ 2e^{2x} & -3e^{-3x} \end{vmatrix} = -3e^{2x}e^{-3x} - 2e^{2x}e^{-3x} = -5e^{-x} \neq 0$$

Since $W \neq 0$ everywhere, they are **linearly independent**. ✓

Example 2: Are $y_1 = \sin 2x$ and $y_2 = \sin x \cos x$ linearly independent? Using the identity

$$\sin 2x = 2 \sin x \cos x,$$

we get

$$y_2 = \frac{1}{2} \sin 2x = \frac{1}{2} y_1.$$

Therefore, the functions are **linearly dependent**.

Note $y_2 = \frac{1}{2} \sin 2x = \frac{1}{2} y_1$. So $y_2 = \frac{1}{2} y_1$ — they are **linearly dependent**.

Check via Wronskian: $y_2' = \cos 2x$, $y_1' = 2 \cos 2x$.

$$W = \sin 2x \cdot \cos 2x - \sin x \cos x \cdot 2 \cos 2x = \cos 2x(\sin 2x - 2 \sin x \cos x) = \cos 2x \cdot 0 = 0 \checkmark$$